

ota-manager shadcn palette — WCAG 2.1 SC 1.4.11 contrast fix + host-render re-proof

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Scope

Applies the WCAG 2.1 SC 1.4.11 (Non-text Contrast, 3.0:1 bar) fixes proposed by docs/research/ota_manager_shadcn_contrast_audit_20260709/AUDIT.md to the shadcn HSL palette in clients/ota-manager/src/index.css (:root = dark theme, .light class = light theme — see AUDIT.md’s theme-naming note, confirmed unchanged here), and re-proves the fix with (1) a real recomputed contrast run and (2) the ota-manager host-render dual-oracle harness (pnpm hostrender). No file outside clients/ota-manager/ and this docs/qa/ directory was modified. No git command was run.

Tokens changed (before → after, both themes)

--border and --input are numerically identical tokens in this codebase in both themes (confirmed in AUDIT.md and in the literal source), so they were changed together as one value per theme.

Theme	Token(s)	Before (HSL / hex)	After (HSL / hex)
dark (:root)	--border, --input	217.2 32.6% 17.5% / #1e293b	217.2 32.6% 39.5434% / #445d86
dark (:root)	--ring	224.3 76.3% 48% / #1d4ed8	224.3 76.3% 48.2524% / #1d4ed9

Theme	Token(s)	Before (HSL / hex)	After (HSL / hex)
dark (:root)	--sidebar-border	217.2 32.6% 14% / #18212f	217.2 32.6% 38.9887% / #435c84
light (.light)	--border, --input	214.3 31.8% 91.4% / #e2e8f0	214.3 31.8% 60.9885% / #7c97bb
light (.light)	--sidebar-border	214.3 31.8% 88% / #d7dfea	214.3 31.8% 59.6316% / #7793b9
light (.light)	--ring	221.2 83.2% 53.3% / #2563eb	unchanged — already PASSED at 5.17:1

All five changed values keep the SAME hue + saturation as the original token (only lightness moved), per the audit's same-hue-family constraint.

Deviation from the task's literal rounded values (honest finding, §11.4.6/§11.4.123)

The task instructions (and AUDIT.md's summary table) quote 1–2-decimal rounded proposed lightness values: dark border/input → 39.5%, dark ring → 48.25%, dark sidebar-border → 39.0%, light border/input → 61.0%, light sidebar-border → 59.6%.

Before committing to those literal values, I recomputed each one at full float precision (not the 2-decimal display rounding AUDIT.md's own report prints). Result: **three of the five rounded values fall a hair BELOW 3.00:1** when recomputed precisely — the 2-decimal display in AUDIT.md's summary table masked this:

Pair	Rounded value → precise ratio	Verdict
dark border/input / background	39.5% → 2.9953189856:1	FAILS (below 3.0)
dark ring / background	48.25% → 2.9997916299:1	FAILS (below 3.0)
light border/input / background	61.0% → 2.9988376807:1	FAILS (below 3.0)

Pair	Rounded value → precise ratio	Verdict
dark sidebar-border / sidebar-background	39.0% → 3.0012338779:1	passes (rounded up)
light sidebar-border / sidebar-background	59.6% → 3.0032501647:1	passes (rounded up)

AUDIT.md itself flagged this risk in its “Notes on the proposed values” section (“A project adopting these would likely round to a slightly higher ratio... for headroom against rounding/rendering variance; the exact binary-search target is reported here for traceability”) — but the literal task instruction quoted the rounded, sub-bar numbers. Per §11.4.6 (no-guessing) and §11.4.123 (rock-solid-proof, never a metadata-only/ display-rounded pass), shipping a value that a full-precision recompute shows is **actually under the bar** would be a contrast-fix that looks right at 2-decimal display but is not real. I used AUDIT.md’s more precise binary-search lightness values (the 4-decimal figures given per-failure in its detailed output, e.g. 39.5434%, 48.2524%, 38.9887%, 60.9885%, 59.6316%) instead of the rounded summary-table numbers, for all five changed tokens (including the two that happened to round up safely, for consistency). This is a same-hue-family, minimal-shift value — identical in spirit to the task’s request, just carried to the precision the real math requires to genuinely clear 3.00:1.

Recomputed ratios — real run, not hand-transcribed

Script: `contrast_recheck.py` in this directory — a copy of the audit’s own `contrast.py` engine with `DARK_TOKENS/LIGHT_TOKENS` updated to the new literal `index.css` values (diff against the original is exactly the 5 token-value edits above; the HSL → sRGB → luminance → contrast-ratio math is byte-identical to the audit’s own, previously-verified-correct engine). Run as `python3 contrast_recheck.py`, Python 3.13.13, **exit code 0**. Full literal stdout captured in `contrast_recheck_output.txt` in this directory.

```
=====
WCAG 2.1 contrast audit -- clients/ota-manager shadcn :root/.light
HSL tokens
SC 1.4.3 Contrast Minimum (text)          bar = 4.5:1
SC 1.4.11 Non-text Contrast (UI/large) bar = 3.0:1
```

```

=====
...
[PASS] dark | border / background |
fg=border(217.2 32.6% 39.5434%) [#445d86] bg=background(222.2 84%
4.9%) [#020817] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] dark | border / card |
fg=border(217.2 32.6% 39.5434%) [#445d86] bg=card(222.2 84% 4.9%)
[#020817] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] dark | input / background |
fg=input(217.2 32.6% 39.5434%) [#445d86] bg=background(222.2 84%
4.9%) [#020817] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] dark | ring / background |
fg=ring(224.3 76.3% 48.2524%) [#1d4ed9] bg=background(222.2 84%
4.9%) [#020817] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] dark | ring / card |
fg=ring(224.3 76.3% 48.2524%) [#1d4ed9] bg=card(222.2 84% 4.9%)
[#020817] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] dark | sidebar-border / sidebar-background |
fg=sidebar-border(217.2 32.6% 38.9887%) [#435c84] bg=sidebar-
background(222.2 84% 3%) [#01050e] | ratio=3.00:1 bar=3.0:1
class=ui
...
[PASS] light | border / background |
fg=border(214.3 31.8% 60.9885%) [#7c97bb] bg=background(0 0% 100%)
[#ffffff] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] light | border / card |
fg=border(214.3 31.8% 60.9885%) [#7c97bb] bg=card(0 0% 100%)
[#ffffff] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] light | input / background |
fg=input(214.3 31.8% 60.9885%) [#7c97bb] bg=background(0 0% 100%)
[#ffffff] | ratio=3.00:1 bar=3.0:1 class=ui
[PASS] light | ring / background |
fg=ring(221.2 83.2% 53.3%) [#2563eb] bg=background(0 0% 100%)
[#ffffff] | ratio=5.17:1 bar=3.0:1 class=ui
[PASS] light | ring / card |
fg=ring(221.2 83.2% 53.3%) [#2563eb] bg=card(0 0% 100%) [#ffffff]
| ratio=5.17:1 bar=3.0:1 class=ui
[PASS] light | sidebar-border / sidebar-background |
fg=sidebar-border(214.3 31.8% 59.6316%) [#7793b9] bg=sidebar-
background(210 40% 98%) [#f8fafc] | ratio=3.00:1 bar=3.0:1
class=ui
=====

```

SUMMARY: 38 pairs checked, 38 PASS, 0 FAIL

=====

No AA failures found among the audited pairs.

Full output (all 38 pairs, both themes) is in `contrast_recheck_output.txt` in this directory — every ratio is the literal computed output, nothing hand-computed or estimated.

Result: all 10 previously-failing pairs now PASS. 0 FAIL across all 38 audited pairs in both themes. All 28 previously-passing text pairs (SC 1.4.3) are untouched and remain at their original PASS ratios (no foreground/background text-pair token was edited by this fix).

No fix skipped

Every one of the 5 proposed same-hue-family token changes (10 failing pairs, since `border==input`) was applied — none was skipped. `--border` and `--input` could not be treated separately (per the task's own instruction and per AUDIT.md's finding that they are numerically identical tokens, and `* { border-color: hsl(var(--border)); }` applies the same value globally including to the card's own border), so there is no way to fix the interactive input border independently of the card's outer border in this codebase's current token structure — fixing one fixes/changes both. `--sidebar-border` was also fixed (not skipped): host-rendering (see below) shows the resulting divider color is a clearly visible, coherent slate-blue tone in both themes — same hue family as every other border in the design, not a jarring or mismatched color — so there was no honest reason to leave it un-fixed. `--ring` in the light theme was correctly left unchanged (it already passes at 5.17:1; no failure was reported for it).

Host-render re-proof (§11.4.170 dual-oracle, pnpm hostrender)

Ran the existing ota-manager host-render harness (`clients/ota-manager/visual/run-all.mjs`, invoked via `pnpm hostrender` which runs `hostrender:build` then the harness) against the LoginPage screen — the harness that already exists in this codebase covering the card/input/ring border tokens this fix changed. The harness regenerates its own baseline + identical re-render + mutated (golden-bad) PNGs on every invocation (self-contained per-run, not diffed against a stale git-tracked golden), so simply re-running it after the CSS edit produces fresh evidence reflecting the new border/ring colors.

Command: `pnpm hostrender` (from `clients/ota-manager/`). Real exit code: **0**. Console summary (verbatim):

```
==== $11.4.170 HOST-RENDER DUAL-ORACLE SUMMARY ====
```

```
[light]
```

```
  image-diff  golden-good : ratio=0.0000%  -> PASS (matches
baseline)
  image-diff  golden-bad  : ratio=1.4688%  -> FLAGGED (regression
caught)
  ocr golden-good          : ALL PRESENT
  ocr golden-bad          : FLAGGED missing "OTA Manager"
(missing=["OTA Manager"])
  layout (baseline)       : OK (no collapse/clip/offscreen/overlap)
  layout (golden-bad)     : FLAGGED collapsed submit ["submit:
COLLAPSED (398.0x0.0)"]
```

```
[dark]
```

```
  image-diff  golden-good : ratio=0.0000%  -> PASS (matches
baseline)
  image-diff  golden-bad  : ratio=1.5501%  -> FLAGGED (regression
caught)
  ocr golden-good          : ALL PRESENT
  ocr golden-bad          : FLAGGED missing "OTA Manager"
(missing=["OTA Manager"])
  layout (baseline)       : OK (no collapse/clip/offscreen/overlap)
  layout (golden-bad)     : FLAGGED collapsed submit ["submit:
COLLAPSED (398.0x0.0)"]
```

```
---- analyzer self-validation ----
```

```
  image-diff analyzer sound : true
  layout   analyzer sound   : true
  ocr      analyzer sound   : true
```

```
OVERALL: PASS
```

Self-validation verdict (§11.4.107(10)): all three oracles (image-diff, layout, OCR) are SOUND for both themes — each one's golden-good input PASSES AND its golden-bad (deliberately mutated/suppressed) input is correctly FLAGGED. This is real evidence the oracles genuinely depend on the rendered pixels/text, not a rubber-stamp.

Fresh baseline PNGs were visually inspected (`docs/qa/20260709-ota-manager-hostrender/baselines/login-dark.png` and `login-light.png`, both regenerated by this run) — both themes render the card, title,

description, email/password fields, and submit button with no overlap, clipping, or collapse; the card and input borders and the (dark theme) focus-ring color are now clearly, legibly visible against their respective backgrounds (previously near-invisible at ~1.2–1.4:1), and the overall look is coherent — a visibly more accessible but not visually “wrong” or off-brand result in either theme.

Honest coverage gap: the existing harness renders only the LoginPage screen, which has no sidebar. It therefore does not host-render-verify the `--sidebar-border` divider color change directly (that token is only used in `sidebar.tsx/app-layout.tsx`, dashboard-shell screens outside this harness’s current screen set). Extending the harness to a sidebar-containing screen was out of this task’s scope (reuse the existing harness, do not build a new one); this gap is recorded per §11.4.3 rather than silently assumed clean. The `--sidebar-border` contrast fix is proven by the real recomputed ratio (3.00:1, above) but not by host-rendered pixels in this pass.

Build + unit tests

- `pnpm build (vite build)`: **exit 0**. 1865 modules transformed, CSS bundle 47.16 kB emitted, built in 2.68s. No new build error introduced by the CSS edit (bundler size warning is pre-existing and unrelated to this change).
- `node_modules/.bin/vitest run`: **9 test files passed (9), 36 tests passed (36)**, Duration 12.71s. No test failed or regressed by this change (no unit test asserts a specific token hex/HSL value, so none was expected to be sensitive to this fix — confirmed by the clean run).

Files changed / left in the working tree

- `clients/ota-manager/src/index.css` — the 5 token edits (both themes), with inline comments citing the before-value, ratio, and this evidence file.
- `docs/qa/20260709-ota-manager-shadcn-contrast-fix/EVIDENCE.md` — this file.
- `docs/qa/20260709-ota-manager-shadcn-contrast-fix/contrast_recheck.py` — the recomputation script (audit’s engine + updated token tables).

- docs/qa/20260709-ota-manager-shadcn-contrast-fix/contrast_recheck_output.txt — full literal stdout of the recheck run (38/38 PASS).
- docs/qa/20260709-ota-manager-hostrender/** — regenerated by pnpm hostrender (baselines/rerender/mutated PNGs, diff PNGs, OCR text, bounds JSON, results.json, harness-src/) — this is the harness's own pre-existing evidence directory from a prior stream's §11.4.170 work; re-running it was the mandated re-proof step, not a new directory of my own.
- clients/ota-manager/dist/** — rebuilt output from pnpm build (build artifact; not itself a hand-authored deliverable).

No file under dashboard/, design-systems/, server/, or clients/ota-manager/src/styles/opendesign-tokens.css was read, edited, or otherwise touched by this work. No git command was run.

Sources verified

Same as docs/research/ota_manager_shadcn_contrast_audit_20260709/AUDIT.md:

- WCAG 2.1 Success Criterion 1.4.11 Non-text Contrast: <https://www.w3.org/WAI/WCAG21/Understanding/non-text-contrast.html>